

Swiss Cooperation with the Standard Nuclear oliGARCHy

A Theory of Monetary, Diplomatic, and Strategic Integration

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Abstract

This paper develops a theory of possible cooperation between Switzerland and the Standard Nuclear oliGARCHy (SNoG). It is not an official Swiss position, an SNoG treaty, or a prediction of diplomatic conduct. Switzerland is treated as a sovereign state with its own constitutional, monetary, diplomatic, and security institutions; SNoG is treated as the formal model defined by its own assumptions and propositions. The central thesis is that cooperation is best represented as a multidimensional state-space problem linking monetary institutions, finance, information, diplomacy, strategic stability, welfare, and security. Mathematics is therefore used principally to formalize the architecture rather than to over-derive it.

The paper ends with “The End”

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1 Introduction

The proposed Swiss–SNoG relationship should first be understood as a theory of cooperation and only subsequently, if competent sovereign authorities independently choose to do so, as a possible legal instrument. Under the Vienna Convention on the Law of Treaties, a treaty is an international agreement between States in written form and governed by international law, irrespective of its particular designation [3].

Swiss neutrality has both an international-law dimension and a broader foreign- and security-policy dimension; official Swiss material also identifies good offices and bridge-building as part of its international role [4, 5]. The Swiss National Bank (SNB) is an independent central bank whose mandate centers on price stability while taking economic developments into account [6].

SNoG is treated differently. In this paper, its claims are propositions internal to the SNoG framework unless independently established. In particular, claims concerning convergence, nuclear diplomacy, induction, diplomatic participation, or systemic inevitability are not presented here as established facts about the international system.

2 Epistemic Foundation: Representation and Reality

The monetary paper motivating this work emphasizes the distinction

$$\text{nominal representation} \neq \text{underlying reality}. \quad (1)$$

A currency unit, financial valuation, statistical observation, diplomatic communiqué, treaty clause, or machine-learning output is an abstraction. Its usefulness depends on what information it preserves and discards [2, 8, 9].

Let the underlying institutional state be

$$X_t = (R_t, M_t, F_t, D_t, S_t, W_t, I_t), \quad (2)$$

where R denotes real resources, M monetary conditions, F financial conditions, D diplomacy, S strategic conditions, W welfare, and I information.

An observable representation is

$$Y_t = \Phi(X_t) + \varepsilon_t. \quad (3)$$

If Φ is many-to-one, then

$$\Phi(X_t) = \Phi(X'_t) \not\Rightarrow X_t = X'_t. \quad (4)$$

Representation Theorem. A non-injective representation cannot uniquely identify the underlying institutional state. Therefore, no single financial, statistical, diplomatic, or computational observable should be treated as a complete representation of the relationship.

This is the paper's basic epistemic constraint.

3 Institutional State Space

A coupled system can be written as

$$X_{t+1} = F(X_t, U_t, \varepsilon_t), \quad (5)$$

where U_t contains monetary, financial, diplomatic, and strategic actions.

For two institutional actors,

$$X_t = (X_t^{CH}, X_t^{SNoG}, X_t^I), \quad (6)$$

where CH denotes Switzerland and I the interaction environment. A cooperation protocol is a constrained policy correspondence

$$U_t \in \mathcal{U}(X_t). \quad (7)$$

The architecture is therefore

$$\text{Cooperation} \neq \text{institutional absorption}, \quad (8)$$

$$\text{Swiss sovereignty} \neq \text{SNoG institutional identity}. \quad (9)$$

4 Swiss Institutional Constraints

Official Swiss sources describe permanent neutrality as a legally recognized status and as a broader foreign-policy instrument associated with credibility, mediation, and good offices [4, 5]. The SNB's mandate is anchored in the Federal Constitution and the National Bank Act, with price stability as its principal monetary-policy objective [6].

Consequently, a theoretical cooperation architecture should preserve

$$\text{Swiss sovereignty} \neq \text{SNoG sovereignty}, \quad (10)$$

$$\text{SNB mandate} \neq \text{SNoG monetary policy}, \quad (11)$$

$$\text{Swiss neutrality} \neq \text{automatic strategic alignment}. \quad (12)$$

These are design constraints, not necessarily obstacles.

5 Monetary and Financial Architecture

Money is an institutional and informational system rather than merely a physical object [2]. Let

$$M_t = (P_t, i_t, E_t, L_t, \pi_t), \quad (13)$$

where P is the price level, i the policy rate, E the exchange rate, L liquidity, and π inflation.

A stylized monetary transmission mechanism is

$$i_t \rightarrow L_t \rightarrow (C_t, I_t) \rightarrow (Y_t, P_t), \quad (14)$$

$$E_t \rightarrow (P_t, Y_t, F_t). \quad (15)$$

The SNB describes its monetary-policy strategy in terms of a definition of price stability, a conditional inflation forecast, and implementation through monetary instruments including the policy rate and, where necessary, foreign-exchange interventions [6, 7].

Thus cooperation need not imply monetary unification:

$$\mathcal{M}_{coop} = \mathcal{M}_{CH} \times \mathcal{M}_{SNoG} \times \mathcal{M}_{interface}. \quad (16)$$

For financial assets,

$$r_{j,t+1} = \frac{P_{j,t+1}}{P_{j,t}} - 1, \quad (17)$$

and a state-dependent expected excess return can be represented as

$$E_t[r_{j,t+1} - r_{f,t}] = \lambda_t^\top \beta_{j,t}. \quad (18)$$

For a multidimensional portfolio,

$$P_t = \sum_{i=1}^m \sum_{j=1}^{n_i} w_{ij,t} P_{ij,t}, \quad \sum_{i=1}^m \sum_{j=1}^{n_i} w_{ij,t} = 1. \quad (19)$$

The risk-premium vector may depend on economic, diplomatic, and strategic states:

$$\lambda_t = \Lambda(X_t^{economic}, X_t^{diplomatic}, X_t^{strategic}). \quad (20)$$

6 Econometrics and Statistics

A basic panel representation is

$$Y_{it} = \alpha_i + \beta^\top Z_{it} + \gamma^\top X_t + u_{it}. \quad (21)$$

A state-space representation is

$$X_{t+1} = AX_t + BU_t + \eta_t, \quad (22)$$

$$Y_t = CX_t + \nu_t. \quad (23)$$

The principal empirical difficulties include endogeneity of diplomatic events, simultaneous financial reactions, selection into negotiations, stale private valuations, heterogeneous institutional rights, non-synchronous observations, and partially unobservable strategic information. Hence causal interpretation requires explicit identification assumptions rather than correlation alone.

7 Physics and Chemistry

Physics contributes dimensional discipline. A physical substrate and a monetary claim should not be assigned identical dimensions merely because one is carried by the other:

$$[V_{monetary}] \neq [V_{substrate}]. \quad (24)$$

Dimensional consistency is necessary for a coherent model but does not by itself establish economic meaning.

Chemistry contributes the distinction between composition and function: systems with similar components may have different structures and therefore different properties. Institutional analogies should be interpreted similarly.

8 Biology and Resilience

A resilient biological system maintains internal stability while exchanging matter, energy, and information with its environment. An analogous institutional condition is

$$\text{Resilience} = f(\text{redundancy}, \text{adaptation}, \text{feedback}, \text{repair}). \quad (25)$$

The desired relationship is therefore robust interdependence rather than total dependence.

9 Psychology and Psychiatry

Diplomatic agents process incomplete signals through priors, heuristics, emotions, strategic incentives, and institutional narratives. A Bayesian representation is

$$p(H \mid D) \propto p(D \mid H)p(H). \quad (26)$$

Repeated communication and verification can reduce uncertainty without requiring complete information.

Psychiatry should be used cautiously. Strategic disagreement, unusual behavior, or divergent beliefs are not evidence of psychiatric disorder. Responsible diplomacy requires distinguishing institutional incentives and cognitive bias from clinical diagnosis.

10 Philosophy and Epistemology

The map–territory distinction provides a general rule: representation is useful precisely because it compresses reality, and compression entails information loss. If X is the underlying state and $\Phi(X)$ its representation, one may conceptualize information loss as

$$\mathcal{L}_{info} = I(X; X) - I(X; \Phi(X)). \quad (27)$$

The objective is not to eliminate abstraction but to preserve sufficient information for the decision at hand.

11 Religion and Symbolic Order

Institutions also depend on symbolic legitimacy. A diplomatic document may possess both instrumental and symbolic value:

$$V_{institution} = V_{instrumental} + V_{symbolic}. \quad (28)$$

Symbolic legitimacy cannot indefinitely substitute for material performance, while material performance without legitimacy may fail to sustain cooperation.

12 Political Science and International Relations

Realism emphasizes power and security; liberal institutionalism emphasizes institutions and repeated cooperation; constructivism emphasizes identity, norms, and legitimacy. The proposed framework treats these as complementary projections:

$$X_t = (Power_t, Institutions_t, Identity_t, Information_t). \quad (29)$$

For repeated interaction, let

$$V_i(C) = u_i(C) + \delta V_i(C). \quad (30)$$

A stylized condition for cooperation is

$$V_i(C) \geq V_i(D), \quad (31)$$

where D denotes defection. This is a model condition, not a prediction about any particular actor.

13 Geopolitics and Diplomacy

International currency importance depends on trade, reserves, finance, settlement infrastructure, liquidity, legal institutions, and expectations; it should not automatically be equated with complete geopolitical power [2].

Represent the international system as a weighted graph:

$$G_t = (V, E, W_t). \quad (32)$$

A stylized currency-utility function is

$$U_c = aL_c + bS_c + cR_c - dK_c, \quad (33)$$

where L is liquidity, S settlement utility, R reserve utility, and K transaction cost.

Diplomacy can therefore be understood partly as maintaining reliable institutional edges in a strategic network.

14 Nuclear Diplomacy and Strategic Stability

Within SNoG, nuclear diplomacy is modeled as a systemic-risk architecture incorporating deterrence, reassurance, communication, bargaining, and escalation management.

Let q_t denote catastrophic-escalation probability. A stylized objective is

$$\max [W_t + D_t + K_t - \omega q_t], \quad \omega > 0, \quad (34)$$

subject to

$$q_t \leq q_{\max}. \quad (35)$$

Swiss neutrality and good offices provide a potentially relevant theoretical interface for communication and mediation, but this does not imply acceptance of any particular SNoG policy [4, 5].

15 Welfare Economics and Warfare Economics

A welfare function can be written as

$$\mathcal{W}_t = W(C_t, L_t, S_t, I_t). \quad (36)$$

Military capability depends on real resources:

$$Q_t^{mil} = F(K_t, L_t, E_t, M_t, T_t), \quad (37)$$

where K is capital, L labor, E energy, M materials, and T technology. Thus

$$\text{money} \neq \text{real resources}. \quad (38)$$

Financial capacity can command resources, but cannot make the resource constraint disappear.

16 Computer Science, Algorithms, and AI

A conceptual feedback controller is:

INPUT:

```
Swiss institutional state
SNoG model state
financial observations
diplomatic observations
strategic-risk observations
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REPRESENT:

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construct X_t
estimate uncertainty
preserve provenance
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UPDATE:

estimate latent state
update Bayesian beliefs
detect structural changes

OPTIMIZE:

evaluate cooperation constraints
evaluate welfare
evaluate systemic risk
evaluate strategic stability

ACT:

choose an admissible action

VERIFY:

observe outcomes
compare predictions with observations
preserve an audit trail

REPEAT.

A multi-objective learning system may use

$$J = J_{\text{economic}} + J_{\text{diplomatic}} + J_{\text{security}} + J_{\text{welfare}} - \lambda J_{\text{risk}}. \quad (39)$$

The objective must remain explicit because

$$J_{\text{model}} \neq J_{\text{state}} \neq J_{\text{society}}. \quad (40)$$

A practical architecture is therefore a hybrid of state-space estimation, sequential modeling, causal inference, machine learning, and human institutional oversight.

17 Proposed Cooperation Algorithm

Algorithm: Swiss--SNoG Cooperation Assessment

1. Establish institutional identities and legal competences.
2. Separate Swiss institutional facts from SNoG model assumptions.
3. Construct a common state representation.
4. Identify observables, latent variables, and uncertainty.
5. Test compatibility with Swiss law and neutrality.
6. Test compatibility with international law.
7. Evaluate monetary and financial interactions.
8. Evaluate diplomatic and strategic interactions.
9. Compute welfare and systemic-risk measures.
10. Stress-test adverse scenarios.
11. Identify information omitted by the proposed representation.
12. Specify verification, dispute-resolution, and communication mechanisms.
13. Do not convert the theory into a treaty unless competent authorities independently determine that a legal instrument is appropriate.

18 Formal Compatibility Conditions

Let $\mathcal{S}$ denote Swiss institutional constraints, $\mathcal{N}$ SNoG constraints, and $\mathcal{I}$ international-law constraints. Define

$$\mathcal{C} = \mathcal{S} \cap \mathcal{N} \cap \mathcal{I}. \quad (41)$$

A coherent cooperation theory requires

$$\mathcal{C} \neq \emptyset. \quad (42)$$

Proposition. If two systems possess compatible feasible sets, cooperation has non-negative expected continuation surplus, and catastrophic-risk constraints remain satisfied, then a cooperative state is theoretically admissible.

Proof. Non-empty feasibility establishes an admissible state. Non-negative continuation surplus prevents immediate defection from strictly dominating continuation in the model. The risk constraint excludes cooperative paths that violate the specified catastrophic-risk bound. Hence an admissible cooperative state exists. Existence does not imply adoption by either sovereign actor. $\square$

19 Information Security and Verification

Let information be partitioned as

$$I = I_{public} + I_{shared} + I_{restricted} + I_{classified}. \quad (43)$$

A disclosure rule should maximize useful information while limiting security cost:

$$\max [I(\Theta; D) - \lambda Risk(D)]. \quad (44)$$

This is the information-theoretic form of diplomatic prudence.

20 Institutional Architecture

Table 1: Proposed dimensions of Swiss–SNoG cooperation

Dimension	Swiss-side reference	Theoretical SNoG interface
Sovereignty	Constitutional institutions	Institutional independence
Neutrality	Permanent neutrality and good offices	Strategic-risk containment
Monetary policy	SNB price-stability mandate	Monetary-system compatibility
Finance	Financial-market infrastructure	Systemic-risk and capital architecture
Diplomacy	Mediation and international relations	Nuclear diplomacy and communication
Information	Legal and institutional confidentiality	Verification and controlled disclosure
Security	National defence and resilience	Deterrence and catastrophic-risk control
Welfare	Economic prosperity and stability	Systemic welfare objective
Legal form	Constitutional and international-law constraints	Model rules, if applicable

21 Vector Representation

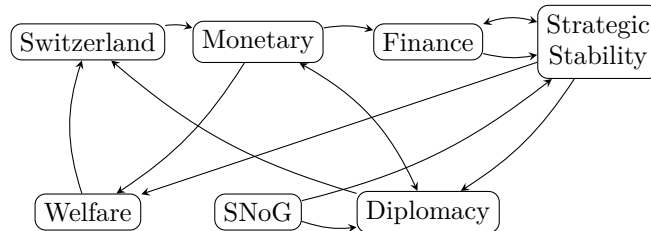


Figure 1: Vector representation.

The arrows denote proposed feedback relationships, not uniquely identified causal effects.

22 Treaty Design as Information Compression

If competent authorities ever decide to negotiate a legal instrument, it can be represented abstractly as

$$T = \Psi(X^{Swiss}, X^{SNoG}, X^{International}). \quad (45)$$

A legal instrument cannot encode the entire relationship. The design problem is therefore

$$\min_{\Psi} \mathcal{L}_{institutional} \quad \text{subject to} \quad \mathcal{L}_{legal} \leq \bar{L}, \quad \mathcal{L}_{security} \leq \bar{S}. \quad (46)$$

A robust instrument would address scope, definitions, institutional competence, information sharing, financial interfaces, diplomatic communication, strategic safeguards, verification, dispute resolution, suspension, termination, amendment, transparency, and confidentiality.

23 The Swiss–SNoG Interface: Necessary Conditions

The preceding sections describe the relevant dimensions. A stronger theory requires explicit interface conditions. Let a cooperation state be

$$Z_t = (A_t, R_t, I_t, G_t, Q_t, W_t), \quad (47)$$

where A_t denotes institutional autonomy, R_t reciprocity, I_t information quality, G_t governance compatibility, Q_t strategic risk, and W_t welfare.

Define the admissible set

$$\mathcal{A} = \{Z_t : A_t \geq \underline{A}, R_t \geq \underline{R}, I_t \geq \underline{I}, G_t \geq \underline{G}, Q_t \leq \overline{Q}\}. \quad (48)$$

The thresholds are not numerical constants at this stage. They represent institutional requirements that would have to be specified by competent authorities or by a future empirical study.

Autonomy Condition. Cooperation must not require either party to surrender institutional competences that its constitutional or legal order reserves to itself. In particular,

$$A_{CH} \geq \underline{A}_{CH}, \quad A_{SNoG} \geq \underline{A}_{SNoG}. \quad (49)$$

Reciprocity Condition. A sustainable interface requires sufficiently symmetric obligations relative to the benefits and risks allocated to each participant:

$$B_{CH} - C_{CH} \approx B_{SNoG} - C_{SNoG}, \quad (50)$$

where B denotes expected benefits and C expected costs. Exact equality is neither necessary nor generally desirable; the relevant condition is that neither side is structurally required to bear uncompensated obligations.

Information Condition. Cooperation requires enough verified information to distinguish compliance from non-compliance:

$$I(\Theta; D) \geq I_{\min}, \quad (51)$$

where Θ is the underlying state and D the disclosed data.

Risk Condition. The relationship must not create an unacceptable increase in systemic or strategic risk:

$$\Delta Q_{coop} = Q_{post} - Q_{baseline} \leq \overline{\Delta Q}. \quad (52)$$

Governance Condition. Decision rights must be attributable to identifiable institutions:

$$g : \mathcal{D} \rightarrow \mathcal{H}, \quad (53)$$

where $\mathcal{D}$ is the set of decisions and $\mathcal{H}$ the set of competent human institutions. An undefined decision-maker is an institutional failure mode.

24 Compatibility Matrix

The theory can be represented as a constraint matrix. Let

$$C = \begin{bmatrix} C_{\text{legal}} & C_{\text{neutrality}} & C_{\text{monetary}} \\ C_{\text{financial}} & C_{\text{diplomatic}} & C_{\text{security}} \\ C_{\text{information}} & C_{\text{welfare}} & C_{\text{governance}} \end{bmatrix}. \quad (54)$$

Each entry represents a compatibility test rather than a score. For example, C_{monetary} asks whether the proposed interface preserves the independent legal mandate of the relevant monetary institution; C_{security} asks whether cooperation remains within agreed strategic-risk limits.

A useful classification is:

Table 2: Interface tests

Test	Question	Failure signal
Legal	Is the proposed action within the competent legal authority?	Ultra vires action
Neutrality	Does the action preserve applicable neutrality obligations?	Military alignment incompatible with neutrality
Monetary	Does it preserve the SNB's statutory framework?	External direction of monetary policy
Financial	Are settlement and systemic-risk channels controlled?	Unbounded contagion
Diplomatic	Are communication and representation rights explicit?	Ambiguous authority
Security	Does cooperation reduce rather than create unacceptable strategic risk?	Higher catastrophic exposure
Information	Can claims be independently verified?	Unverifiable commitments
Welfare	Are material effects assessed beyond nominal indicators?	Proxy/reality confusion
Governance	Is responsibility assigned to competent institutions?	Accountability gap

25 Equilibrium Interpretation

Suppose each side chooses a cooperation intensity $x_i \in [0, 1]$. Let

$$U_i(x_i, x_j) = B_i(x_i, x_j) - C_i(x_i, x_j) - R_i(x_i, x_j). \quad (55)$$

A cooperative equilibrium is a pair (x_{CH}^*, x_{SNoG}^*) satisfying

$$x_i^* \in \arg \max_{x_i \in [0, 1]} U_i(x_i, x_j^*), \quad i \in \{CH, SNoG\}, \quad (56)$$

subject to

$$Z_t \in \mathcal{A}. \quad (57)$$

The theory does not require maximal cooperation. It requires an admissible equilibrium in which cooperation is sufficiently valuable, sufficiently verifiable, and sufficiently safe.

26 Stress Testing

A serious cooperation theory should survive adverse scenarios. Define scenario s with probability p_s and welfare-security payoff V_s . Expected performance is

$$E[V] = \sum_s p_s V_s. \quad (58)$$

But expected value alone is insufficient where catastrophic losses are highly nonlinear. Introduce a tail-risk functional $\mathcal{T}$:

$$\mathcal{R} = E[V] - \lambda \mathcal{T}(V). \quad (59)$$

Relevant stress scenarios include:

1. a severe international financial shock;
2. a sudden monetary or exchange-rate disturbance;
3. a diplomatic crisis involving a participant;
4. a disputed interpretation of a cooperation clause;
5. a cyber or information-system disruption;
6. an escalation event involving nuclear-armed states;
7. a major change in the international security environment.

The key question is not whether every scenario can be prevented. It is whether the interface contains credible mechanisms for detection, communication, containment, and recovery.

27 Verification, Audit, and Falsifiability

The theory should distinguish a proposition from its test. Let a model claim be

$$H : F(X_t, \theta) = 0. \quad (60)$$

A corresponding empirical test requires observable implications

$$Y_t \sim P_\theta(Y_t | X_t). \quad (61)$$

This principle is especially important for SNoG propositions. A mathematical result conditional on axioms is a theorem of the model; it becomes an empirical claim only when its assumptions and observable implications are connected to the real world.

A future implementation should therefore maintain an audit record:

$$\mathcal{A}_t = (\text{claim, source, assumption, observation, inference, decision}). \quad (62)$$

This creates provenance from evidence to institutional action.

28 Diplomatic Time and Sequencing

The relationship should also be treated as a sequential game. Let the stages be

$$\text{recognition} \rightarrow \text{exploration} \rightarrow \text{verification} \rightarrow \text{limited cooperation} \rightarrow \text{review} \rightarrow \text{expansion or termination}. \quad (63)$$

This sequencing matters because information acquired at an earlier stage changes the feasible set at a later stage:

$$\mathcal{C}_{t+1} = \mathcal{C}_t \cap \mathcal{I}(D_t), \quad (64)$$

where $\mathcal{I}(D_t)$ denotes the constraints implied by verified information.

The result is a formal version of a basic diplomatic principle: access should not be confused with consent, and consent should not be confused with permanent commitment.

29 Termination and Reversibility

A cooperation architecture should contain a controlled exit mechanism. Let S_t denote the cooperation state. A termination rule can be represented as

$$\tau = \inf \{t : Z_t \notin \mathcal{A}\}. \quad (65)$$

Termination need not imply hostility. A reversible framework can define suspension, consultation, remediation, and eventual re-entry:

$$\text{cooperation} \rightarrow \text{suspension} \rightarrow \text{review} \rightarrow \begin{cases} \text{restoration,} \\ \text{modification,} \\ \text{termination.} \end{cases} \quad (66)$$

Reversibility can itself reduce strategic risk because actors need not treat every disagreement as an irreversible confrontation.

30 The Core Theorem of the Interface

Swiss–SNoG Compatibility Theorem. Suppose that (i) the institutional identities of Switzerland and SNoG remain distinct, (ii) the cooperation state satisfies the legal, neutrality, monetary, information, governance, and strategic-risk constraints, (iii) each participant receives non-negative expected cooperation surplus, and (iv) compliance can be verified with sufficient information. Then the proposed cooperation interface is internally coherent as a theoretical institutional arrangement.

Proof. Condition (i) prevents institutional merger from being an implicit requirement. Condition (ii) places the relationship inside the admissible set $\mathcal{A}$. Condition (iii) ensures that continuation is not strictly dominated by the modeled alternative. Condition (iv) permits the parties to distinguish compliance from non-compliance. Therefore an internally coherent cooperation state exists. The theorem establishes model coherence, not political feasibility, legal adoption, or empirical success. $\square$

31 Implication for a Future Legal Instrument

If competent authorities ever decide to translate the theory into an international agreement, the legal instrument should be treated as a controlled interface rather than as a complete representation of the relationship.

The design principle is therefore

$$\text{Legal instrument} \subset \text{Institutional relationship.} \quad (67)$$

The treaty or agreement should encode only those commitments that can be clearly attributed, lawfully undertaken, verified, administered, reviewed, and, where necessary, suspended or terminated. The larger relationship will necessarily contain political judgment, informal communication, institutional learning, and contextual information that no document can fully encode.

32 Limits

Four limitations are central. First, SNoG assumptions are not automatically empirical facts. Second, mathematical existence does not imply political adoption. Third, statistical association does not establish causality. Fourth, institutional cooperation cannot eliminate strategic uncertainty.

33 Conclusion

The proposed Swiss–SNoG relationship is best represented as a multidimensional cooperation problem:

$$\boxed{\text{Resources} \rightarrow \text{Representation} \rightarrow \text{Institutions} \rightarrow \text{Diplomacy} \rightarrow \begin{matrix} \text{Strategic} \\ \text{Stability} \end{matrix} \rightarrow \text{Welfare} \rightarrow \text{Feedback}} \quad (68)$$

The monetary perspective supplies the distinction between nominal representation and underlying reality. Finance supplies risk pricing and capital allocation. Econometrics supplies identification and uncertainty. Physics supplies dimensional discipline. Chemistry supplies the distinction between composition and function. Biology supplies resilience. Psychology supplies belief updating and bounded cognition. Psychiatry supplies a caution against pathologizing disagreement. Philosophy supplies the map–territory distinction. Religion supplies symbolic legitimacy. Political science and international relations supply institutions, power, norms, and repeated games. Welfare economics supplies the social objective. Warfare economics supplies real-resource constraints. Computer science and AI supply state representation, feedback control, algorithms, and machine-assisted inference.

The final principle is

$$\boxed{\text{Cooperation} = \begin{matrix} \text{compatible} \\ \text{institutions} \end{matrix} + \begin{matrix} \text{credible} \\ \text{information} \end{matrix} + \text{reciprocity} - \begin{matrix} \text{uncontrolled} \\ \text{systemic} \\ \text{risk} \end{matrix}} \quad (69)$$

Whether any formal agreement should ever exist is a separate sovereign and legal question. The theory’s task is to make that question structurally intelligible.

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Glossary

Bayesian updating

Revision of beliefs after observing new information.

Cooperation interface

Institutional mechanism connecting distinct systems without requiring institutional merger.

Dimensional consistency

Requirement that mathematical quantities possess compatible dimensions.

Neutrality

Switzerland's legally recognized status and associated foreign-policy practice, including good offices.

Risk premium

Expected compensation for bearing systematic or otherwise priced risk.

SNoG

Standard Nuclear oliGARCHy, treated here as the user's formal systemic framework.

State space

Set of variables sufficient to describe the relevant condition of a dynamical system.

Systemic risk

Risk that interactions among components generate consequences beyond an isolated failure.

Treaty

International agreement meeting the relevant requirements of international law.

True risk premium

State-dependent compensation for bearing economically relevant risk, as used in the user's asset-pricing framework.

The End